

4.18 CW Normal & Binomial Distributions Solutions
12 March 2026

1. Bakery croissants

a) $E(x) = 10 \cdot 0.15 = 1.5$

b) $P(x=0) = 0.85^{10} = 0.19687...$
 ≈ 0.197

c) $P(x=0,1) = \text{binomCDF}(10, 0.15, 0, 1)$
 $= 0.5443...$
 ≈ 0.544

d) $P(x=2) = \text{binomPDF}(10, 0.15, 2)$
 $= 0.275897...$
 ≈ 0.276

e) Duplicate 0.544

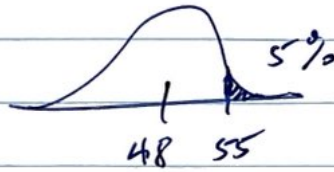
2. a) $\text{norm}(\text{~~145~~, 12, 160, \infty, 145, 12})$
 $= 0.10565... \approx 0.106$

b) $\text{normCDF}(130, 155, 145, 12)$
 $= 0.69202... \approx 0.692$

c) $\text{invNorm}(0.8, 145, 12) = 155.099...$
 ≈ 155

3. a)

$$Z = \text{invNorm}(0.95, 0, 1) \\ = 1.64485...$$



$$Z = \frac{55 - 48}{\sigma} = 1.64485...$$

$$\sigma = \frac{7}{1.64485} = 4.2557... \\ \approx 4.26$$

$$b) \text{Norm}(-\infty, 42, 4.26) = 0.079288... \\ \approx 0.0793$$

$$c) \text{binom CDF}(20, 0.0793, 0, 2) \\ = 0.791712... \\ \approx 0.792$$

$$d) P(X < 42 | X < 52) = \frac{P(X < 42)}{P(X < 52)} \\ = \frac{0.0793}{0.82637...} \\ = 0.095948... \\ \approx \cancel{0.0960} \\ \approx 0.0959$$